Table S1. Physicochemical parameters and analytical data of major elements of the spring waters.

No.	Site	Date	Longi-tude	Latitude	Tempre-ture(℃)	PH	con-ductiv-ity	Na^+ (mg/L)	K^+ (mg/L)	Mg^{2+} (mg/L)	Ca^{2+} (mg/L)	F^- (mg/L)	Cl^- (mg/L)	SO_4^{2-} (mg/L)	NO_3^- (mg/L)	CO_3^{2-} (mg/L)	HCO_3^- (mg/L)	Chemical type	87Sr/86Sr
		2017.3.9						760.39	11.87	41.52	366.67	2.63	789.30	1564.65	0.00	0.00	34.96		
W1	MD	2017.3.29	99.6268	26.4357	34	7.19	4590	790.01	13.32	35.20	352.78	2.93	825.98	1538.49	4.29	0.00	663.48	SO4Cl-NaCa	0.708718
		2018.4.22						804.75	13.15	36.66	375.90	3.08	814.33	1580.14	2.60	0.00	719.29		
		2021.5.24						735.20	14.30	31.54	421.78	2.66	764.47	1504.11	0.05	0.00	712.18		
		2015.3.19						232.36	34.80	14.12	70.36	5.87	17.21	151.68	2.96		627.82		
		2016.4.20						263.95	8.70	0.84	63.78	4.20	14.93		4.31	0.00	580.66		
		2017.2.28						201.07	28.98	13.67	62.35	4.58	16.59	142.00	3.52	0.00	655.41		
W2	NJ	2017.3.29	99.9877	26.2537	78	8.24	1200	201.07	28.98	13.67	62.35	4.58	16.59	142.00	3.52	0.00	655.41	HCO3SO4-Na	0.708718
		2018.3.27						213.98	31.53	15.01	69.90	5.04	16.22	144.26	3.98	0.00	708.58		
		2019.5.16						216.78	31.06	13.86	68.26	5.12	16.70	147.21	3.15	166.40	331.35		
		2021.5.26						209.98	31.93	15.49	86.40	4.26	20.12	149.80	4.24		640.22		
		2017.3.29						640.11	36.47	7.18	45.65	7.43	26.69	514.88	0.43	0.00	1287.30		
		2018.3.27						640.39	37.21	6.56	47.66	8.73	26.41	535.86	2.75	54.12	1180.96		
W3	EYXX	2019.5.16	99.9913	26.2519	72	6.62	2800	653.37	36.21	6.37	40.66	9.40	26.61	543.11	2.53	496.45	1274.03	HCO3SO4-Na	0.712455
		2021.5.24						638.76	36.75	7.40	60.55	8.03	30.34	550.42	0.08	0.00	1417.75		
		2015.3.19						293.99	44.30	7.76	46.63	6.80	18.46	210.18					
		2016.4.18						295.16	47.46	8.72	46.16	6.32	17.19	210.99	1.20	0.00	598.04		
		2017.2.28						250.43	38.83	7.64	43.97	5.42	14.84	187.35	0.00	0.00	664.15		
W4	HYS	2017.3.29	99.9926	26.2392	74	6.91	1505	273.13	42.69	8.49	43.15	5.68	16.26	200.82	0.00	0.00	672.22	HCO3SO4-Na	0.708887
		2018.3.27						274.43	42.20	8.39	50.70	7.05	16.49	209.20	1.29	27.72	660.94		
		2019.5.16						282.26	42.53	8.37	59.17	6.86	16.91	219.08	1.47	191.84	318.77		
		2021.5.25						280.36	43.52	9.26	67.12	6.59	17.87	233.87	0.06	0.00	778.36		
		2015.3.19						165.80	17.90	3.41	38.22	6.82	12.34	238.91			241.23		
		2017.2.28						164.74	17.15	3.11	40.81	6.05	11.82	244.45	0.00	0.00	272.25		
W5	XSK	2017.3.20	100.0252	26.0843	92	8.13	962	169.58	18.21	3.46	36.63	6.10	11.94	239.30	0.00	0.00	269.56	HCO3SO4-Na	0.707998
		2018.3.28						171.14	18.63	3.27	33.29	7.86	11.50	241.03	0.08	32.34	213.38		
		2021.5.25						174.91	18.95	3.42	47.56	7.40	14.39	255.71	0.05	31.73	282.06		

Table S1. *Cont.*

No.	Site	Date	Longi-tude	Latitude	Tempre-ture(°C)	PH	con-ductiv-ity	Na^+ (mg/L)	K^+ (mg/L)	Mg^{2+} (mg/L)	Ca^{2+} (mg/L)	F^- (mg/L)	Cl^- (mg/L)	SO_4^{2-} (mg/L)	NO_3^- (mg/L)	CO_3^{2-} (mg/L)	HCO_3^- (mg/L)	Chemical type	87Sr/86Sr
		2016.4.18						158.87	16.57	8.98	54.36	3.88	16.39	42.19	2.75	0.00	531.98		
		2017.2.26						176.06	17.28	9.50	39.76	4.16	21.05	48.28	0.68	0.00	544.50		
		2017.3.28						155.72	15.23	9.34	54.77	3.66	15.39	41.67	0.00	0.00	542.48		
W6	SB	2018.3.26	99.9204	25.8047	70	7.43	930	153.90	14.81	8.43	55.76	4.73	15.20	39.00	0.00	38.28	501.24	HCO3SO4-Na	0.715199
		2019.5.17						154.09	14.40	8.43	53.28	4.36	15.70	39.55	1.09	121.02	290.11		
		2021.5.24						151.82	14.82	9.25	69.64	3.80	18.21	43.84	0.10	0.00	603.00		
		2015.3.20						274.34	18.76	6.33	66.05	6.71	11.21	304.79			556.69		
		2016.4.17						286.12	18.69	6.12	59.82	5.94	14.05	324.98	0.83	0.00	563.27		
		2017.2.26						311.74	20.70	6.38	45.54	6.32	13.53	379.93	0.00	0.00	571.39		
W7	QG	2017.3.28	100.1822	25.5697	78	7.4	1480	286.36	19.36	6.03	58.41	5.49	11.17	317.61	0.00	0.00	563.99	HCO3SO4-Na	0.715374
		2018.3.26						285.92	19.37	5.83	60.39	6.54	10.90	329.86	0.00	39.60	518.68		
		2019.5.17						154.09	14.40	8.43	53.28	4.36	12.49	339.03	1.09	121.02	290.11		
		2021.5.24						290.82	20.09	6.07	75.08	5.79	12.68	341.97	0.06	0.00	665.04		
		2017.2.26						103.64	4.22	1.30	12.64	5.82	5.20	98.35	0.00	0.00	160.66		
		2017.3.28						92.16	3.87	1.00	10.98	5.19	4.31	84.44	0.00	0.00	155.96		
W8	JXQ	2018.3.26	100.1299	25.5621	63	8.21	471	96.60	3.85	0.99	12.93	6.21	3.76	82.89	1.91	0.00	186.54	HCO3SO4-Na	0.718426
		2019.5.17						92.96	3.45	1.07	12.63	5.81	3.97	84.70	0.16	43.32	72.70		
		2021.5.23						89.83	3.39	0.99	17.21	5.32	6.82	82.45	0.00	0.00	189.42		
		2015.3.16						964.44	19.18	8.20	60.26	5.68	57.31	695.64			1765.93		
		2016.4.3						984.46	19.34	8.13	60.32	5.15	58.14	719.51		0.00	1826.65		
		2017.2.25						1033.66	22.36	8.50	24.23	5.33	65.31	826.28	0.76	0.00	1937.34		
W9	WBS	2018.4.20	100.323	25.1706	46	7.32	3840	960.82	18.46	8.43	58.73	5.33	57.29	720.09	5.58	136.32	1710.92	HCO3SO4-Na	0.711761
		2019.5.17						969.15	17.79	6.95	57.05	5.83	56.51	738.65	4.87	394.68	1174.42		
		2021.5.23						980.09	22.31	9.11	80.87	4.74	62.68	760.20	0.44	0.00	2278.83		
W10	XQC	2021.5.23	99.7904	25.4829	56			151.82	14.82	9.25	69.37	3.80	18.21	43.84	0.10	0.00	603.00	SO_4-Ca	0.708728

Table S2. Analytical data of trace elements on hot spring water sample .

Hot Spring	Date	Li	Be	Ba	B	Sr	Ti	V	Cr	Mn	Fe	Co	Ni
		ug/L											
MD	2021.5.24	217	0.683	29.9	614	9662	3.77	10.5	1.65	139	0.023	1.6	23.5
NJ	2021.5.26	1145	3.62	96.8	2029	988	10.5	6.35	1.04	9.67	0.014	0.3	5.85
HYS	2021.5.25	1631	6.22	107	2385	1364	11.8	0.97	1.22	75.5	0.137	0.231	4.68
EYXX	2021.5.24	795	23.7	50.5	4.17	2396	9.58	1.04	1.48	137	0.051	0.353	5.01
XSK	2021.5.25	1172	0.929	51.3	852	881	11.4	0.605	0.532	32	0.045	0.142	4.28
JXQ	2021.5.25	232	0.131	8.32	215	127	6.02	0.402	0.442	9.79	0.045	0.057	1.31
SB	2021.5.23	573	1.18	85.5	70.7	254	12	0.725	0.716	488	0.012	0.223	4.08
XCQ	2021.5.23	200	0.725	33.2	8.73	12534	4.87	0.288	0.635	32.4	0.039	1.52	33.2
QG	2021.5.24	1324	1.54	66	1483	975	8.19	0.554	0.867	117	0.062	0.295	6.49
WBS	2021.5.23	2047	0.886	53.2	18330	2754	6.29	1.98	3.57	34.1	0.027	0.412	8.06

Hot Spring	Date	Cu	Zn	Mo	Cd	Sb	Tl	Pb	Th	U	Ag	Sn	Al
		ug/L											
MD	2021.5.24	25.8	218	1.04	0.049	0.464	0.087	0.117	<0.002	0.405	0.003	0.434	2.03
NJ	2021.5.26	17.5	80.2	1.13	0.014	14.1	1.21	0.141	<0.002	6.74	0.003	0.369	14.6
HYS	2021.5.25	18.8	75.1	0.538	0.041	4.84	1.06	0.16	0.009	0.203	0.003	0.536	10.9
EYXX	2021.5.24	23.5	83.9	5.24	0.029	6.04	0.271	0.169	0.003	0.259	0.003	0.426	42.3
XSK	2021.5.25	15.8	62.9	15	0.033	2.53	0.525	1.06	<0.002	0.078	<0.002	0.588	5.71
JXQ	2021.5.25	13.7	50	1.74	0.018	0.861	0.01	0.131	<0.002	0.529	<0.002	0.425	13.1
SB	2021.5.23	15.4	79.7	0.168	0.066	0.241	0.03	0.147	<0.002	0.159	0.004	0.495	6.09
XCQ	2021.5.23	18.9	106	1.66	0.038	19.7	0.053	0.15	0.002	0.386	0.005	0.356	36.9
QG	2021.5.24	21.1	136	0.255	<0.002	12.2	0.201	0.797	<0.002	0.139	<0.002	0.525	6.47
WBS	2021.5.23	19.7	74.2	0.226	0.008	51.9	0.01	0.221	<0.002	0.139	0.002	0.556	6.08